

Reconstructing turbulence at Reynolds numbers we cannot verify: the anchor rules as they now stand, and the experiment now running

A plain-language account. Prepared 8 August 2026, from the measured record in the repository (docs/anchor_derived_rules.md and the committed result files). Every number cited here was measured; where a rule rests on an assumption rather than a measurement, the text says so explicitly.

1. The problem in one paragraph

We own a diffusion model that turns a coarse simulation of two-dimensional turbulence into a fine one. It was trained at one flow regime, Reynolds number one thousand. We want to use it, and refined versions of it, at other Reynolds numbers - flows that are more (or less) turbulent than anything it saw in training. The defining constraint is that at those target regimes we possess only the coarse observations. There is no fine-resolution reference to compare against, because producing one is exactly the expense the model exists to avoid. Every choice made at a target regime - whether to fine-tune at all, which training checkpoint to keep, and how deeply to run the sampler - must therefore be made from the coarse data alone, guided by what we learned at the two regimes whose truth we do hold.

2. What information we allow ourselves

Reynolds numbers five hundred and one thousand are the calibration regimes: we hold their full fine-resolution data and may use it freely there. Everything else is out of distribution: the pipeline may see only a coarse sixty-four by sixty-four observation of those flows. To make this auditable rather than promised, the coarse observations were written once to separate files (one per regime, eleven in total), and it was proven that these files carry exactly the information the old code path used: the reconstructed inputs are bit-identical and the coarse spectra agree to machine precision. Downstream code reads those files and nothing else, so a fine-resolution field is not merely unused at a target regime - it is not present to be leaked.

3. The anchor: a target you can build without the truth

The fine-tuning reward and all selection decisions need a target: what should the energy spectrum of the true flow look like? The anchor is that target. It is one curve, predicted energy at every wavenumber from one to one hundred and twenty-eight, and it is assembled in three zones of decreasing trust. Below wavenumber six it is essentially measured, read directly from the coarse data with a small correction. Between wavenumbers six and thirty it is fitted: the coarse spectrum, corrected by a transfer function, is fitted with a standard four-parameter spectral form. Above wavenumber thirty it is pure extrapolation - the fitted formula evaluated where no observation exists.

Four constants inside that procedure cannot be measured at the target and are frozen from the calibration regimes: the spectral slope and cutoff-sharpness priors, the law by which the dissipation wavenumber grows with Reynolds number, and the transfer function relating coarse to fine spectra. Each anchor also stores a fingerprint of the coarse data it was built from, and the trainer refuses to start if the fingerprint does not match the data in front of it - a guard added after a stale anchor silently invalidated an early sealed test. A second pre-flight rejects data whose frame spacing is incompatible with the model's training convention.

Honesty about accuracy: even at the calibration regimes the anchor does not match the true spectrum exactly, because it is fitted to coarse data only, through a deliberately constrained four-parameter family. Measured at

Reynolds five hundred it runs about six percent hot over the scoring band. At out-of-distribution regimes an audit of the deployed two-reference procedure found the bias grows with Reynolds number, up to roughly nineteen percent hot. This drift is the weakest joint in everything that follows, and it is declared rather than hidden.

4. The blind score, and why healthy is not one

The one number computable at any regime without truth is the blind score: the energy the reconstruction contains, summed over wavenumbers ten to ninety-five, divided by the energy the anchor predicts over the same range. A score of zero point eight means the output carries eighty percent of the fine-scale energy the anchor claims the true flow has. A genuinely healthy model does not read one point zero on this instrument: the anchor is slightly hot, and even a verified-excellent model does not match a fitted analytic curve shell by shell. Where health could be verified against truth, the healthy models read zero point seven eight four (Reynolds one thousand) and zero point eight three five (Reynolds five hundred). Empirically, cells that do read near one point zero are over-restorers, often with badly damaged structure.

5. How the healthy readings were obtained: the grid

At each calibration regime we built a full grid: nine sampler depths (from a single shallow chain of twelve steps up to a seven-chain cascade of two hundred steps) crossed with thirteen models (the untuned base plus twelve checkpoints of a fine-tuning run), every cell graded against ground truth on held-out sequences - one hundred and twenty-six cells in all. Three findings shaped the rules.

First, the sampler depth dominates the checkpoint. Depth moves the restored energy three to four times more than checkpoint choice does, and depth is free at inference while a checkpoint costs a four-hour training run. Second, different regimes need opposite depths: Reynolds five hundred is only fully restored by the five-chain cascade, while Reynolds one thousand overshoots badly there and wants the three-chain one. No single frozen depth can serve both, which is why depth must be chosen per regime. Third, the total amount of restored energy alone cannot define health. Twelve cells at Reynolds five hundred sit within ten percent of the perfect energy total while their structural quality spans from excellent to unusable - one cell has the right band total while matching the true spectrum at essentially no wavenumber. Health was therefore defined by two measurements jointly: energy within ten percent of truth, and, among those, the best structural placement. Under that definition the two regimes' healthy readings nearly agree (zero point seven eight four and zero point eight three five), where a naive energy-only definition had made them look far apart.

A finding worth stating on its own: at Reynolds five hundred the best cell overall is the untuned base model at the five-chain depth. Fine-tuning helps nothing there and damages structure. The rule below therefore always includes the untuned base among its candidates, so 'do not fine-tune' is a permitted conclusion.

6. The selection rule, as engraved

At a target regime, after the two data pre-flights pass, the procedure is:

Step 1. Compute the base-deficit statistic: how much of the anchor's demanded fine-scale energy the untuned base already delivers. If it is high (as at Reynolds five hundred, where it read zero point seven four), fine-tuning is likely unnecessary; the statistic also fixes the training temperature by the frozen bucket rule if a fine-tune is run.

Step 2. Reconstruct the coarse observations once with the untuned base model (twenty denoising steps). This shared reconstruction is the starting point for every candidate.

Step 3. Score every candidate cell - each pairing of a model (base or any saved checkpoint) with each of the nine sampler depths - by running its cascade on the shared reconstruction and computing the blind score on the anchor's own source sequences.

Step 4. Shortlist the cells whose blind score falls inside the band from zero point seven seven to zero point eight five - the interval covering both verified healthy readings plus measurement noise.

Step 5. Apply the structure guard: compute each shortlisted cell's map of where its fine-scale energy sits, and correlate it with the same map of the shared base reconstruction. Discard any cell whose correlation falls more than zero point two zero below the shallowest depth's value. This catches candidates that have moved their fine-scale energy away from anything the observation supports. It is a veto only - it is never maximised, because it systematically flatters shallow, conservative candidates.

Step 6. Choose the survivor whose blind score is nearest zero point eight one. If several are within noise of each other, prefer the shallower depth, then the earlier checkpoint, then the untuned base - conservatism along exactly the axes the score cannot see.

Step 7. Apply the low-frequency patch: splice the base reconstruction's spectrum below the crossover wavenumber (chosen by a frozen, truth-free rule) with the selected output's spectrum above it. This repairs the large-scale erosion that training and depth cause, restoring the low band to within about one percent at no cost to the fine scales.

Also recorded, but never acted upon: the physics residual of the output relative to the base reconstruction's, the raw low-frequency ratio, and the margin between the winner and the runner-up. These are diagnostics for later audit.

7. What this costs

Selection is one cascade pass per cell over roughly sixty coarse inputs. The full grid of one hundred and seventeen cells costs about one and a half hours on the current hardware, roughly forty percent of one fine-tuning run; with the measured monotonicity of depth used to bracket rather than sweep, and the checkpoint list thinned, it falls to about half an hour. When the base-deficit statistic says fine-tuning is unnecessary, the whole procedure is nine cells and a few minutes. Both costs scale with the model in the same way fine-tuning does, so the ratio survives a larger network.

8. Stated limitations

- The band was calibrated at the two owned regimes, both at or below the base model's training Reynolds number. Using it at higher Reynolds numbers is extrapolation, and the anchor's known hot drift means a truly healthy cell at high Reynolds number may read below the band's floor. The band is deliberately kept as calibrated; this risk is accepted and recorded rather than guessed at.

- The structure guard assumes the base reconstruction's map of fine-scale activity points at the structures the observation determines. That assumption was validated where truth exists (correlation with true placement of zero point nine six and zero point seven four at the two calibration regimes) and holds in audits of the generated research regimes, but at a genuine deployment regime it is unverifiable. Its failure is bounded: the guard can only discard, never promote.
- The two verified healthy readings are two points. They cannot distinguish a constant from a slow drift; they only rule out the large regime-dependence an earlier, flawed analysis suggested.
- Structural placement itself - whether the fine detail is in the right places - remains unmeasurable without truth. Above roughly ten times the training Reynolds number, previous work found placement collapses even when the energy totals are right; the rule cannot see that, and no truth-free instrument we have tested can.

9. The experiment now running

The question under test: did the second calibration regime actually improve the rule? Two arms are run, identical in every respect except what the calibration knew. The first arm knows only Reynolds one thousand: its set-point is zero point seven eight four and its band is the narrow interval around that single measurement. The second arm knows both regimes: set-point zero point eight one, band zero point seven seven to zero point eight five. Both arms run the full selection rule at nine generated research regimes spanning Reynolds fifteen hundred to ten thousand - the full grid where a native fine-tune exists (Reynolds two thousand), and the base model across all nine depths elsewhere.

Selection in both arms is entirely truth-free. Afterwards - and only afterwards - every depth of the base model and every cell either arm picked is graded against the held-out fine-resolution data of those research regimes, which we possess precisely so that such audits are possible. That grading yields, per regime: the choice each arm made, what each choice truly delivered, and the best choice available in hindsight. If the two arms pick identically everywhere, the second calibration regime bought confidence but not decisions. If they differ, the audit says which calibration was right and by how much. Either outcome also answers the practical question of whether buying ground truth at further reference regimes is worth the expense.

10. The constants, in one place

Constant	Value	Origin
Set-point (both regimes)	0.810	midpoint of the two verified healthy readings
Acceptance band	0.770 - 0.850	covers both readings plus measurement noise
Set-point (single-regime arm)	0.784	Reynolds one thousand calibration alone
Structure-guard margin	0.20	keeps both verified winners; rejects all cells with true placement be
Tie window on the blind score	0.02	the score's measured noise
Health gate at calibration	energy within 10% of truth	wide enough to expose the structural spread; then best placement
Sampler depths considered	9 (12 to 200 steps)	extended until the optimum fell in the interior at both calibration reg

Every number above is frozen before any out-of-distribution use. Changing any of them after seeing out-of-distribution results would invalidate the exercise; if one is changed, the change and its reason are to be recorded in the provenance log first.